



香 港 大 學

THE UNIVERSITY OF HONG KONG

Progress Report

2026 W15 - W16

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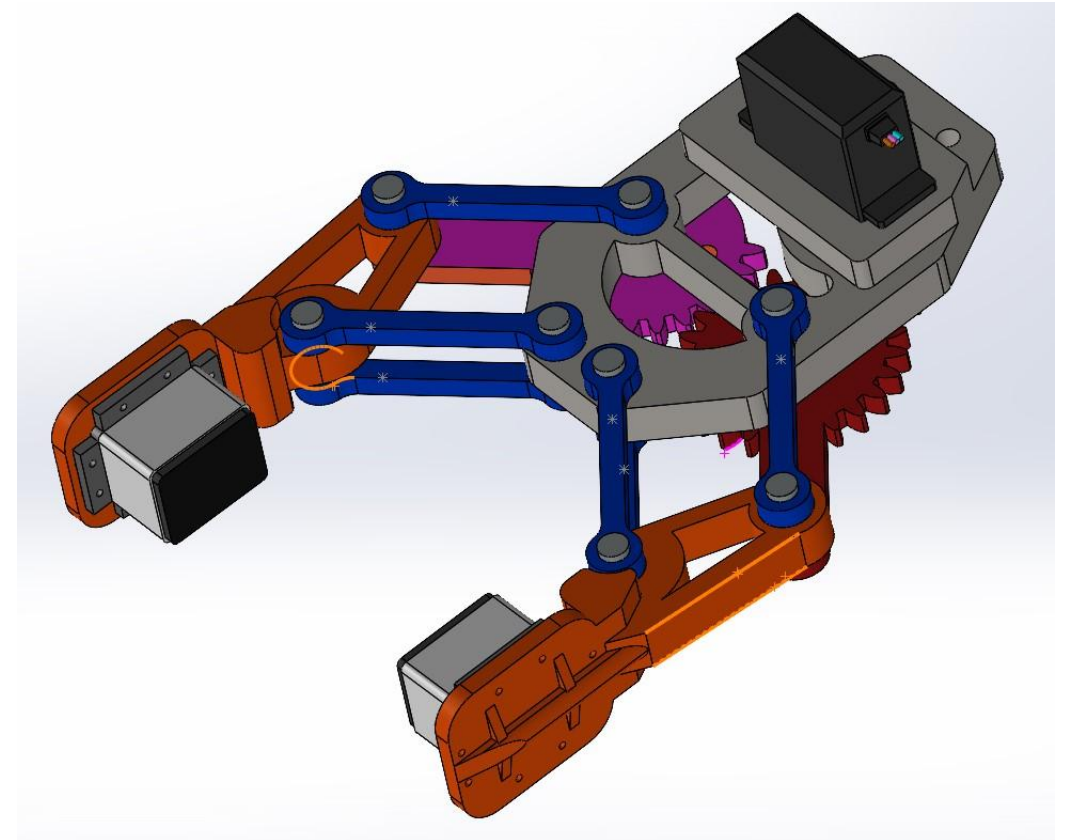
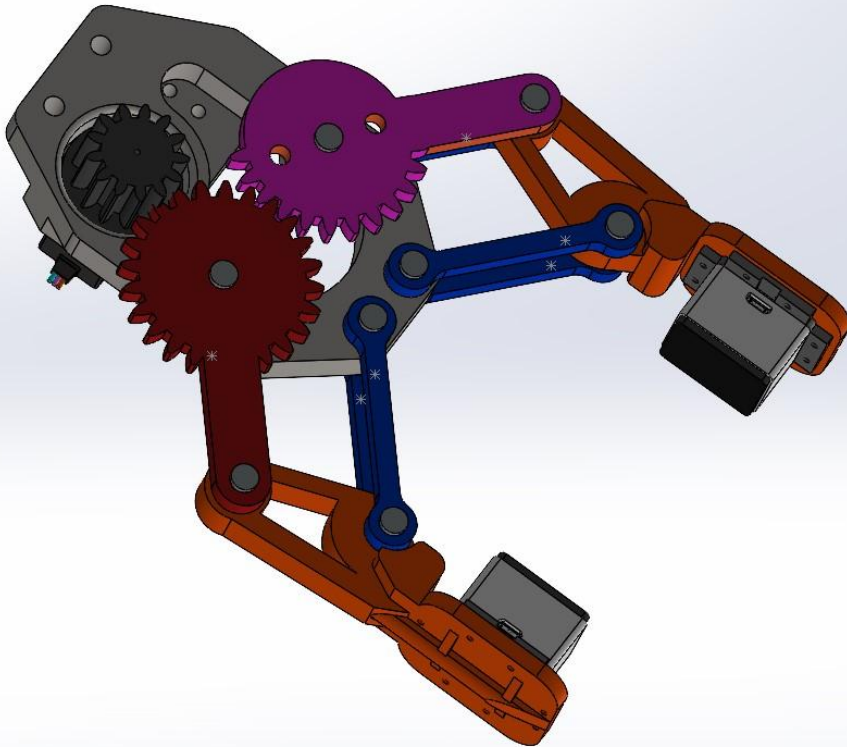
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1.1 Gripper Design & Tactile Feedback Loop

Aiming to provide a reliable physical validation platform for the application and performance testing of tactile sensors, I designed a robotic gripper.



1.1 Gripper Design & Tactile Feedback Loop

The Principles of Visuo-Tactile Closed-Loop Control:

- **Phase 1: Pre-Calibration & Initial Approach**

- I. **Calibration Force Setting:** Upon system initialization, a safe and basic "calibration grasping force" is preset as the initial target.
- II. **Initial Closure:** The controller sends an open-loop command to the MG996R servo, driving the gripper to close until the silicone surface just contacts the object and reaches an initial, slight force state.

- **Phase 2: Photometric Perception & 3D Reconstruction**

- I. **Capturing Illumination Distortion:** As the silicone deforms under pressure, the light intensity and shadows generated by the internal LED array change subtly. The OV5640 camera captures these high-resolution image signals in real-time.
- II. **Image to Tactile Force:** The Raspberry Pi uses photometric stereo algorithms to perform a 3D geometric reconstruction of the contact surface based on light intensity changes, thereby accurately calculating the current physical contact force and stress distribution.

1.1 Gripper Design & Tactile Feedback Loop

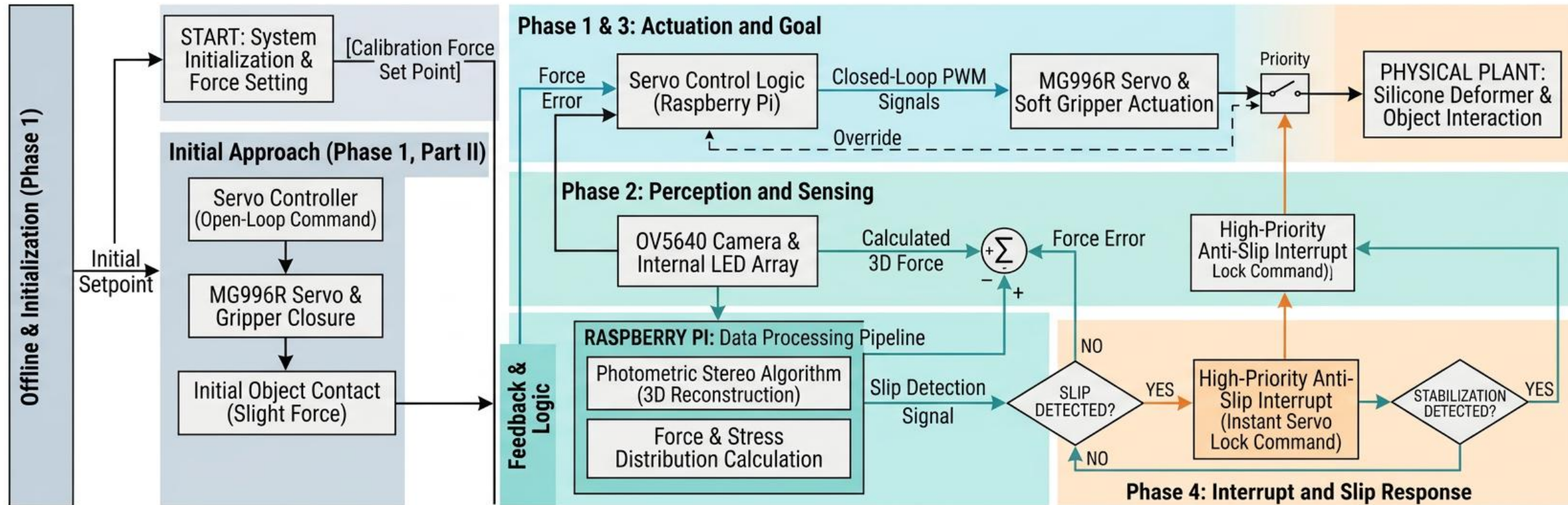
● Phase 3: Error Calculation & Dynamic Fine-Tuning

- I. **Real-Time Correction:** The Raspberry Pi calculates the error between the "3D-reconstructed real-time force" and the "calibration force" at a high frequency.
- II. **Flexible Adaptation:** If the current force is less than the calibration force, it fine-tunes the PWM signal to slowly increase pressure; if the force is too high, it slightly opens the gripper. Through this millimeter-level dynamic adjustment, it achieves a perfect "soft grasp" on fragile or unknown-hardness objects.

● Phase 4: Slip Warning & Transient Response

- I. **Shear Force Monitoring:** During the 3D reconstruction, if dynamic downward displacement of the contact surface texture is detected (i.e., a sudden change in shear force), the system immediately determines that the object is slipping.
- II. **Anti-Slip Interrupt:** It triggers the highest-priority control, instantaneously increasing the servo's closing angle to lock the object, and automatically switches back to the regular force fine-tuning loop once stabilized.

1.1 Gripper Design & Tactile Feedback Loop



1.2 Refining the BOM details

Part	No.	Item Name	Item Category	Specifications	Quantity	Unit price	Source/Link
Robotic Gripper	1	Flat Head Screw	Standard Parts	M6*40	10	Total 5 CNY	Taobao
	2	Flat Head Self-tapping Screw	Standard Parts	M4*12	50	Total 3 CNY	Taobao
	3	Hex Nut	Standard Parts	M6	50	Total 5.8 CNY	Taobao
	4	Plain Washer	Standard Parts	M6*12*1	100	Total 5.5 CNY	Taobao
	5	Flat Head Screw	Standard Parts	M2*14	50	Total 3.3 CNY	Taobao
	6	Hex Nut	Standard Parts	M2	50	Total 3 CNY	Taobao
	7	Plain Washer	Standard Parts	M2*4.5*0.5	150	Total 4.5 CNY	Taobao
	8	3D Printing Service	Custom Parts	SLA / PLA (1 Set)	1	150 CNY	Taobao
	9	Servo	Standard Size	MG996R (180°)	2	40 CNY	Taobao
	10	Raspberry Pi 4B (4GB) Combo	Electronic Parts	Broadcom BCM2711 / 4GB RAM	1	1300 CNY	Taobao
	11	12V 4A Power Adapter	Electronic Parts	AC to DC 12V 4A	1	60 CNY	Taobao
	12	XL4015 Adjustable Step-down Module	Electronic Parts	DC-DC 5A / 75W	2	20 CNY	Taobao
	13	Dupont Jumper Wires	Electronic Parts	M-M / M-F / F-F, 20cm	30	Total 20 CNY	Taobao
	14	INA219 Power Monitor Module (I2C)	Electronic Parts	I2C Interface, 0-26V	3	5.4 CNY	Taobao
	15						
Tactile Sensor	1	3D Printing Service	Custom Parts	SLA / PLA (1 Set)	3	Total 50 CNY	Taobao
	2	Camera	Electronic Parts	OV5640	3	75 CNY	Taobao
	3	Transparent Acrylic Board	Raw Materials	14mm*14mm*2mm	3	5 CNY	Taobao
	4	High-Transparency Silicone A&B (G40)	Raw Materials	500g	2	40 CNY	Taobao
	5	Translucent Silicone A&B (G5)	Raw Materials	200g	2	20 CNY	Taobao
	6	Silicone Pigment (Black)	Raw Materials	20g	1	18 CNY	Taobao
	7	LED-PCB	Electronic Parts	Ref:PickAndPlace_LED-PCB;BOM_LED-PCB	3	Total 50 CNY	嘉立创
	8	Flat Head Screw	Standard Parts	M2*5	50	Total 4.5 CNY	Taobao
	9	Heat-Set Insert	Standard Parts	M2*5*3	50	Total 5 CNY	Taobao
	10	Flat Head Screw (Bottom to Acrylic)	Standard Parts	M2*6	50	Total 3 CNY	Taobao
	11	Flat Head Screw (Shell to Acrylic)	Standard Parts	M3*30	20	Total 3 CNY	Taobao
	12	Hex Nut (Shell to Acrylic)	Standard Parts	M3	20	Total 3.4 CNY	Taobao
	13	Transparent Acrylic Injection Mold	Custom Parts	Custom Machined	3	Total 20 CNY	Taobao
	14						
Tools & Consumables	1	Adhesive / Glue	Consumables		1	20.8 CNY	Taobao
	2	Tweezers	Hand Tools		1	3.8 CNY	Taobao
	3	Screwdriver & Wrench Set	Hand Tools		1	17 CNY	Taobao
	4	Dust Removal Tape	Consumables		1	7 CNY	Taobao
	5	Transparent Plastic Cups	Consumables		30	Total 18.5 CNY	Taobao
	6	Screw & Nut Organizer Box	Storage		1	20 CNY	Taobao
	7	Hot Melt Glue Gun	Power Tools		1	30 CNY	Taobao
	8	Soldering Iron	Power Tools		1	32 CNY	Taobao
	9	Heat Insulation Pad	Tool Accessories		1	40 CNY	Taobao
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02.

Project Plan for the Next Two Months

Plan (Late April - Early July)

● Phase 1: Hardware Assembly & Preparation (Late April - Mid May)

◆ Key Tasks:

- Complete the silicone molding of the visuo-tactile sensor
- Complete the isolated wiring (high/low voltage) of the gripper.

◆ Critical Challenge:

- Optical-grade Silicone Molding. A vacuum degasser must be used to ensure the silicone is free of bubbles; otherwise, it will severely disrupt subsequent visual feature extraction.

Plan (Late April - Early July)

- **Phase 2: Visuo-Tactile Model Training & Calibration (Mid May - Mid June). . . . [CORE FOCUS]**
- ◆ **Key Tasks:**
 - **Data Collection:** Use the lab's 6D force sensor as Ground Truth. Implement strict timestamp synchronization via Python to collect paired datasets of "OV5640 internal deformation images" and "actual 6D force/torque values" during compression.
 - **OpenCV Preprocessing:** Use OpenCV to calculate the Difference Image between the loaded frame and the initial frame to filter out static backgrounds. Retain only deformation shadows caused by LED illumination distortion, crop the ROI, and convert to Tensors.
 - **PyTorch Training:** Construct a Regression Task neural network on the workstation. Use a lightweight Backbone (e.g., MobileNetV3 or ResNet-18) for feature extraction to output 6D force predictions. Update model weights using MSE Loss and the Adam optimizer.
- ◆ **Critical Challenge:**
 - Spatiotemporal Alignment & Inference Acceleration. Millisecond-level alignment of images and force data is a prerequisite. Meanwhile, the PyTorch model must be exported to ONNX format to ensure real-time control (<50ms per forward pass) on the Raspberry Pi.

Plan (Late April - Early July)

● Phase 3: Closed-Loop System Debugging & Optimization (Late June - Early July)

◆ Key Tasks:

- Deploy the lightweight model to the Raspberry Pi. Execute the complete closed-loop of "visual force measurement -> error calculation -> PWM servo adjustment", conducting flexible grasping and anti-slip response tests.

◆ Critical Challenge:

- Stability of Closed-Loop Response. It requires overcoming the physical latency of the motor and optimizing the trigger sensitivity of the "anti-slip interrupt" mechanism to prevent system oscillation.